A COUNTEREXAMPLE TO A PERIODIC WAITING-TIME ENTROPY BOUND

Harvey Yorke
Valyu AI

Abstract. Maier, Degünther, van der Meer and Seifert conjectured that a waiting-time estimator constructed entirely from the forward protocol bounds the entropy production of a periodically driven Markov network from below. We give an analytic three-state counterexample. The two observed links have symmetric rates $1+\varepsilon\cos t$ and $1+\varepsilon\sin t$, and the hidden link has constant symmetric rate one, with $0<\varepsilon<1$. The stationary distribution remains uniform and the defined Markov entropy production is identically zero. Nevertheless, two reversed event-pair waiting densities have derivative difference ε^2 at zero lag, making the estimator strictly positive. All rates are smooth and uniformly positive. The result concerns the forward-protocol conjecture, not the separate theorem using data from the time-reversed protocol.

Research draft. Exact checks and internal proof reviews are available; author approval, priority assessment and external review remain open.

1. The conjectured bound

Consider a continuous-time Markov chain with periodically driven rates $k_{ij}(t)$, period T, and periodic stationary distribution $p_i(t)$. An observer records a reversal-closed set $\mathcal{L}$ of directed jump labels. Write $\overline{I}$ for the label with opposite direction and $\bar{n}_I$ for the mean occurrence rate of label I. Let $\Psi_{I\to J}(s)$ be the joint density that the next observed label is J and that its delay after an event I is s, averaged over the phase distribution of occurrences of I. Thus

$$\sum_{J\in\mathcal{L}}\int_0^\infty \Psi_{I\to J}(s)\,ds = 1. \tag{1}$$

Individual label-pair densities need not have unit integral.

Maier et al. propose the estimator [3]

$$\widehat{\sigma}_\Psi = \sum_{I,J\in\mathcal{L}} \bar{n}_I \int_0^\infty \Psi_{I\to J}(s)\log\frac{\Psi_{I\to J}(s)}{\Psi_{\overline{J}\to\overline{I}}(s)}\,ds \tag{2}$$

and conjecture $0\le\widehat{\sigma}_\Psi\le\bar{\sigma}$, where

$$\bar{\sigma} = \frac{1}{T}\int_0^T \sum_{i,j} p_i(t)k_{ij}(t)\log\frac{p_i(t)k_{ij}(t)}{p_j(t)k_{ji}(t)}\,dt. \tag{3}$$

These are Eqs. (17), (18), and (3), respectively, of the published article; they correspond to Eqs. (14), (15), and (3) of arXiv:2406.12657v1. Crucially, both waiting densities in (2) are measured under the *same forward protocol*. Only the event labels reverse. The source distinguishes this conjecture from its proved Eq. (16) (arXiv-v1 Eq. (13)), which uses data generated by the reversed protocol.

The published general setup imposes periodic rates and local detailed balance, with no restriction to constant symmetric kinetic prefactors. The counterexample below changes symmetric kinetic rates while keeping all local free-energy differences and affinities zero.

Date: 18 October 2026.

2. A smooth three-state counterexample

Use time units in which the hidden rate and driving angular frequency are both one. Take $T = 2\pi$, $0 < \varepsilon < 1$, and set

$$\begin{aligned} k_{12}(t) = k_{21}(t) = a(t) &= 1 + \varepsilon \cos t, \\ k_{23}(t) = k_{32}(t) = b(t) &= 1 + \varepsilon \sin t, \\ k_{13}(t) = k_{31}(t) &= 1. \end{aligned} \tag{4}$$

Observe $\mathcal{L} = \{12, 21, 23, 32\}$ and leave the edge 13 hidden. All states are ordinary states even under time reversal.

Theorem 1. *For every $0 < \varepsilon < 1$, the network (4) has*

$$\bar{\sigma} = 0 < \widehat{\sigma}_\Psi < \infty.$$

Consequently the upper inequality in the forward-protocol conjecture fails for smooth, strictly positive, periodically driven Markov rates.

Proof. Every instantaneous generator is symmetric. Hence $p_1(t) = p_2(t) = p_3(t) = 1/3$ is the periodic stationary distribution, and each summand in (3) is zero. Positivity of all rates ensures uniqueness of this periodic distribution. The rate ratios are one, satisfying local detailed balance with zero free-energy differences and zero affinities.

Write $\langle q \rangle = (2\pi)^{-1} \int_0^{2\pi} q(t)\, dt$. Since $\langle a \rangle = \langle b \rangle = 1$, each observed directed event has mean rate $\bar{n}_I = 1/3$. Conditional on the occurrence of 12, the phase density is $a(t)/(2\pi)$; conditional on 32, it is $b(t)/(2\pi)$. This event weighting is essential.

After either event the system occupies state 2, whose exits are both observed. Define

$$H(t, s) = \int_t^{t+s} (a(u) + b(u))\, du. \tag{5}$$

The two reversed label-pair densities are therefore exactly

$$\begin{aligned} f(s) := \Psi_{12\to 23}(s) &= \left\langle a(t) b(t+s) e^{-H(t,s)} \right\rangle, \\ g(s) := \Psi_{32\to 21}(s) &= \left\langle b(t) a(t+s) e^{-H(t,s)} \right\rangle. \end{aligned} \tag{6}$$

They satisfy $f(0) = g(0) = \langle ab \rangle = 1$. Differentiation under the phase integral is justified by smoothness on a compact domain. The two survival derivatives cancel, leaving

$$\begin{aligned} (f - g)'(0) &= \langle ab' - ba' \rangle \\ &= \left\langle \varepsilon \cos t + \varepsilon \sin t + \varepsilon^2 \right\rangle = \varepsilon^2 > 0. \end{aligned} \tag{7}$$

Thus $f(s) > g(s) > 0$ on a nonempty interval of positive delays.

Every state has an observed exit rate at least $1 - \varepsilon$. The survival probability of no observed event for a delay s is at most $e^{-(1-\varepsilon)s}$, proving (1). On the product space of label pairs and positive delays, define probability densities

$$P(I, J, s) = \tfrac{1}{4} \Psi_{I\to J}(s), \qquad Q(I, J, s) = \tfrac{1}{4} \Psi_{\overline{J}\to\overline{I}}(s).$$

Both integrate to one because reversal permutes the label pairs. Equal event rates give the exact identity

$$\widehat{\sigma}_\Psi = \frac{4}{3} D_{\mathrm{KL}}(P \| Q) > 0. \tag{8}$$

Strictness follows from (7), since $P \neq Q$ on a set of positive measure.

For completeness, the divergence is finite. From state 2, the only possible next labels are 21 and 23. From states 1 or 3, the only hidden evolution is a rate-one symmetric chain on $\{1, 3\}$, and the next labels are 12 or 32. The allowed pair set is invariant under reversal. The hidden chain's transition probabilities are $(1 \pm e^{-2s})/2$, according to whether its endpoints agree. The

survival weighting multiplies these probabilities by a factor between $e^{-(1+\varepsilon)s}$ and $e^{-(1-\varepsilon)s}$. Put $\delta = 1-\varepsilon$ and $b_* = 1+\varepsilon$. Initial phase and final event-rate factors give, for a pair with hidden endpoint transition probability $q_s = (1 \pm e^{-2s})/2$,

$$\delta^2 e^{-b_* s} q_s \le \Psi_{I\to J}(s) \le b_*^2 e^{-\delta s} q_s.$$

Reversal preserves the hidden endpoint parity, so q_s cancels in the density ratio. For state-2 pairs, (5) gives the same bounds with $q_s = 1$ and doubled decay exponents. Consequently every allowed pair satisfies

$$\left|\log \frac{\Psi_{I\to J}(s)}{\Psi_{\bar{J}\to\bar{I}}(s)}\right| \le 2\log(b_*/\delta) + 4\varepsilon s.$$

The densities are bounded by $b_*^2 e^{-\delta s}$, proving integrability. Impossible pairs vanish in both densities and contribute zero. □

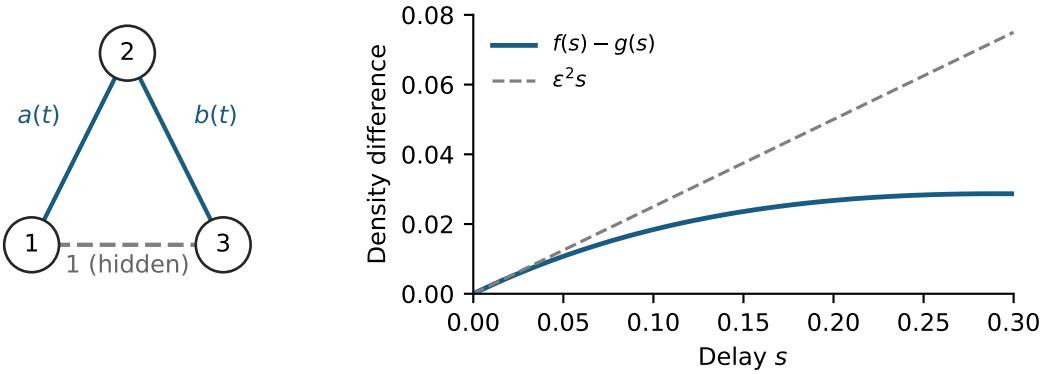


FIGURE 1. Left: the two observed links have symmetric rates $a(t)$ and $b(t)$; the dashed hidden link has symmetric rate one. Right: the difference $f(s) - g(s)$ at $\varepsilon = 1/2$ and its exact tangent $\varepsilon^2 s$ at zero. Periodic quadrature illustrates the densities; the proof uses the analytic identity (7).

3. Interpretation, prior context, and verification

The nonzero estimate records the temporal ordering of kinetic rates. It does not require a probability current: in this family every instantaneous current and every term in (3) vanish. More generally, for positive smooth functions a, b with common mean m, the same construction gives

$$(f-g)'(0) = \frac{\langle ab' - ba' \rangle}{m}.$$

A nonzero signed area enclosed by the periodic rate-plane curve therefore supplies a sufficient condition for this waiting-time asymmetry. This geometric observation explains the mechanism rather than asserting a new general relation between time asymmetry and dissipation.

Waiting-time irreversibility at equilibrium is already known. Hartich and Godec give an equilibrium counterexample based on hidden transitions and explain its kinetic-hysteresis interpretation [1]. Our construction observes actual directed edges; its asymmetry comes from marginalizing the driving phase while retaining the forward protocol in both comparison laws. It is not the first waiting-time false positive.

Protocol reversal is already central to the source's proved bound. Harunari, Fiore and Barato independently study periodically driven waiting-time inference, explicitly use reversed-protocol distributions, and discuss Maier et al.'s forward-protocol proposal as a conjecture [2]. Their kinetic-barrier models are close prior context. The result here is the explicit failure of the stated forward-protocol upper bound. Neither reversed-protocol result is contradicted.

There is also no assertion that a physical autonomous device generating the protocol dissipates no energy. Such a controller is absent from (3). The theorem concerns precisely the Markov entropy production used in the conjecture. It does not identify the total thermodynamic cost of implementing time-dependent rates.

The proof requires no trajectory simulation or parameter search. An independent exact Fourier-polynomial calculation over the rationals verifies

$$f'(0) = -2 - \tfrac{1}{2}\varepsilon^2, \qquad g'(0) = -2 - \tfrac{3}{2}\varepsilon^2.$$

From this directory, the command

```
python3 -B -O verification/check_short_lag.py
```

replays these identities using the Python standard library. `sh build.sh` runs that check and builds both author presentations; `plot_counterexample.py` reproduces the illustration. The exact checker covers the polynomial identities, while the event conditioning, normalization, support, and integrability arguments are proved above and recorded in the separate independent audit. Bounded prior searches and a focused literature review are retained with the working files. They do not establish exhaustive priority.

Research assistance

Valyu DeepResearch supported literature retrieval and source comparison. Coding agents developed the counterexample and analytic argument, implemented the exact verification program, and drafted and revised the manuscript. The exact checks cover the stated Fourier identities; they do not formalize the probability argument or constitute external peer review.

Contributions and AI use

Harvey Yorke directed the project. Coding agents developed the arguments, the verification checkers and the manuscript drafts. Valyu DeepResearch carried out the literature, priority and internal adversarial reviews. These reviews are internal; the manuscript has not undergone external peer review. The author has not yet completed a line-by-line review of every proof; correctness rests on the written proofs and on the checks and any formal verification described in this paper.

References

[1] David Hartich and Aljaž Godec. Comment on "inferring broken detailed balance in the absence of observable currents". arXiv:2112.08978, 2021. Preprint consulted. A comment with the same title and authors was published in Nature Communications 15, 8678 (2024), doi:10.1038/s41467-024-52602-0.

[2] Pedro E. Harunari, Carlos E. Fiore, and Andre C. Barato. Inference of entropy production for periodically driven systems. *Physical Review E*, 110:064126, 2024. arXiv:2406.12792v2.

[3] Alexander M. Maier, Julius Degünther, Jann van der Meer, and Udo Seifert. Inferring kinetics and entropy production from observable transitions in partially accessible, periodically driven Markov networks. *Journal of Statistical Physics*, 191:104, 2024. doi:10.1007/s10955-024-03315-7. Published text consulted; numbering also given for arXiv:2406.12657v1.